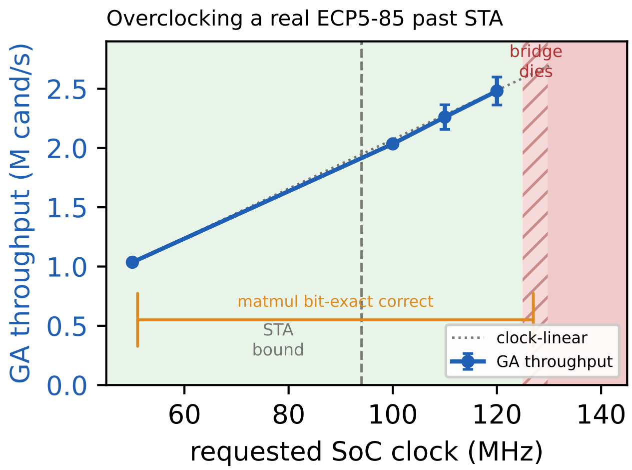

Figure 1: Overclocking the K=16 engine on a real ECP5-85. STA predicts ∼94 MHz, but the design runs correct and throughput scales to 120 MHz; a bit-exact matmul probe is error-free through 128 MHz. The hard limit (∼125–130 MHz, marginal) is the soft-CPU control bridge, not the datapath.

Overclocking Past Timing Closure

An FPGA place-and-route tool reports a maximum clock from *static timing analysis* (STA): a conservative bound at a worst-case process/voltage/temperature corner. It is a prediction, not a measurement — and the LiteX/trellis flow emits a bitstream even when a requested clock *violates* it. We test what the real silicon does past the bound. A per-build clock override (one PLL parameter, exposed as a submission argument) sets the SoC compute-clock; the Ethernet domain is fixed by the RMII PHY, so only the compute fabric is reclocked. We sweep it, build, flash, and run on real ECP5-85 boards, checking functional correctness and throughput.

Silicon beats the model by a wide margin (Figure 1). STA predicts the K=16 engine tops out at ∼94 MHz. On hardware at room temperature it *solves the target and scales* well past that: ∼2.0 M cand/s at 100 MHz (2× the 50 MHz default) and 2.48±0.12 M cand/s at 120 MHz (mean of 6 runs, all solving) — ∼28% beyond STA, a 2.4× gain that *exceeds a full Apple M4* (2.42 M/s) with 16 cores on a ∼$40 part, all correct. To probe correctness directly we ran a bit-exact arithmetic kernel (a 4×4 signed matrix multiply, real MAC logic) over many random inputs against a software reference: it is **bit-exact correct at every clock the board responds to, through 128 MHz** (∼750 multiplies, zero element errors) — ∼36% past STA. The tool's bound is *very* conservative; a large band of real, usable frequency sits above it.

The bottleneck is the control bridge, not the accelerator. The hard limit is not the compute fabric. Around 125–130 MHz the board goes *unresponsive* — the bitstream still programs, but the soft VexRiscv + LiteEth bridge can no longer service packets — and this failure is *marginal and probabilistic*: a 125 MHz build died after 11 transactions while a 128 MHz build completed 150, the run-to-run luck expected right at a timing edge. So the accelerator's arithmetic never produces a wrong answer in the entire usable range; the *control processor* fails first. The actionable implication is that the path to more headroom is a hardened or minimal control plane (e.g. a hardware Etherbone bridge replacing the soft CPU), not a faster datapath.

Implications for evolutionary hardware. The immediate, demonstrated win is concrete: a one-line clock change buys a measured ∼2× throughput at fully correct results, because the conservative STA bound hides that much real margin. The deeper, and we think more interesting, question is whether evolutionary workloads can exploit the regime *past* clean correctness — where the compute itself starts to miss timing and produce silent errors. There is reason to expect they can: an evolutionary search is already a stochastic process that draws mostly-unfit candidates and selects the rare good ones, so a corrupted evaluation is one more noisy sample in a pipeline built to tolerate noise (errors do not accumulate; a wrongly-rejected candidate is replaced by the next draw, a wrongly-promoted one selected against on re-evaluation). This places evolutionary hardware in the lineage of approximate computing and of intrinsic evolvable hardware (Thompson, 1996), which exploited the analog physics of real silicon — here the exploitable physics is timing margin. Yet on this chip we never reach that regime: the engine's throughput stays *clock-linear* with no measurable deficit through 120 MHz (within the ∼5% run-to-run noise; 6 of 6 runs solve at both 110 and 120), and the matmul probe is exact to 128. The compute simply does not err before *the control bridge fails first*. So the error-tolerance hypothesis, though well-motivated, is untested here; confirming it as a usable throughput lever is gated on first removing the bridge bottleneck to expose the datapath's own error regime — a concrete next experiment.

Caveats. Results are one device at one temperature; timing margins and the exact failure clock vary with both. The error-tolerance argument requires *selection* — a pure random search that trusts a single best-fitness read is vulnerable to a false positive from a scoring error, whereas a re-evaluating GA is not. Achieved-Fmax and a timing-met flag, surfaced per build, would make the clean/degraded boundary visible to users rather than inferred.

References

Thompson, A. (1996). An evolved circuit, intrinsic in silicon, entwined with physics. In *Evolvable Systems: From Biology to Hardware (ICES)*, volume 1259 of *LNCS*, pages 390–405. Springer.